

What is a Formula?

A **formula** is an expression that performs calculations using numbers, cell references, and operators.

A formula always starts with an **equal sign (=)**.

Example

Suppose:

A B

10 20

Formula:

=A1+B1

Result:

30

Here:

- = tells Excel this is a formula.
 - A1 and B1 are cell references.
 - + is the operator.
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What is a Function?

A **function** is a predefined formula built into Excel that performs a specific task.

Instead of writing a long formula, Excel provides ready-made functions.

Example:

=SUM(A1:A5)

This adds all numbers from A1 to A5 automatically.

Difference Between Formula and Function

Formula	Function
Created by the user	Built into Excel
Can use operators like +, -, *, /	Performs a specific task
Example: =A1+B1	Example: =SUM(A1:A5)

Parts of a Formula

Example:

=A2*B2

- = → Starts the formula
 - A2 → First value
 - * → Multiplication operator
 - B2 → Second value
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Types of Operators

Arithmetic Operators

Operator	Meaning	Example	Result
+	Addition	=5+3	8
-	Subtraction	=10-4	6
*	Multiplication	=5*4	20
/	Division	=20/5	4
^	Power	=2^3	8
%	Percentage	=500*10%	50

Comparison Operators

These return **TRUE** or **FALSE**.

Operator	Meaning	Example	Result
=	Equal	=A1=B1	TRUE/FALSE
>	Greater Than	=A1>50	TRUE/FALSE
<	Less Than	=A1<50	TRUE/FALSE
>=	Greater Than or Equal	=A1>=50	TRUE/FALSE
<=	Less Than or Equal	=A1<=50	TRUE/FALSE
<>	Not Equal	=A1<>50	TRUE/FALSE

Cell References

Relative Reference

Formula:

=A2+B2

If copied down one row, it becomes:

=A3+B3

Excel changes the references automatically.

Absolute Reference

Formula:

=\$A\$1*B2

\$A\$1 stays fixed when copied.

Useful for tax rates, discounts, or constants.

Mixed Reference

Examples:

=\$A1

Column is fixed.

=A\$1

Row is fixed.

Important Functions

1. SUM()

Adds numbers.

Example:

A

10

20

30

Formula:

=SUM(A1:A3)

Result:

60

2. AVERAGE()

Finds the average.

Formula:

=AVERAGE(A1:A5)

Example:

10, 20, 30, 40

Average:

25

3. MAX()

Returns the largest value.

Example:

=MAX(A1:A5)

Numbers:

20, 60, 80, 30

Result:

80

4. MIN()

Returns the smallest value.

Example:

=MIN(A1:A5)

Result:

20

5. COUNT()

Counts only cells containing numbers.

Example:

A

10

20

ABC

30

Formula:

=COUNT(A1:A4)

Result:

3

6. COUNTA()

Counts all non-empty cells.

Example:

A

10

ABC

30

Hello

Formula:

=COUNTA(A1:A4)

Result:

4

7. COUNTBLANK()

Counts empty cells.

Formula:

=COUNTBLANK(A1:A10)

IF Function

One of the most important functions for Data Analysts.

Syntax

=IF(condition, value_if_true, value_if_false)

Example:

Marks

80

35

Formula:

=IF(A2>=35,"Pass","Fail")

Result:

- 80 → Pass
 - 35 → Pass
 - 20 → Fail
-

SUMIF()

Adds values that meet a condition.

Example:

Department Salary

IT	50000
HR	30000
IT	60000

Formula:

=SUMIF(A2:A4,"IT",B2:B4)

Result:

110000

COUNTIF()

Counts cells that meet a condition.

Example:

=COUNTIF(A2:A10,"IT")

Counts how many employees are in IT.

AVERAGEIF()

Calculates the average for values matching a condition.

Example:

=AVERAGEIF(A2:A10,"IT",B2:B10)

Returns the average IT salary.

LEFT()

Returns characters from the left.

Example:

=LEFT("Bangalore",4)

Result:

Bang

RIGHT()

Returns characters from the right.

=RIGHT("Bangalore",4)

Result:

lore

MID()

Returns characters from the middle.

=MID("Bangalore",5,3)

Result:

(Start at the 5th character and return 3 characters.)

LEN()

Counts the number of characters.

=LEN("Excel")

Result:

5

TRIM()

Removes extra spaces.

Example:

" Ravi Kumar "

Formula:

=TRIM(A1)

Result:

Ravi Kumar

UPPER(), LOWER(), PROPER()

=UPPER("excel")

Result:

EXCEL

=LOWER("EXCEL")

Result:

excel

=PROPER("ravi kumar")

Result:

Ravi Kumar

CONCAT()

Joins text together.

=CONCAT(A2," ",B2)

Example:

First Name = Ravi

Last Name = Kumar

Result:

Ravi Kumar

TODAY()

Returns today's date.

=TODAY()

NOW()

Returns the current date and time.

=NOW()

ROUND()

Rounds a number.

=ROUND(15.678,2)

Result:

15.68

VLOOKUP()

Searches for a value in the first column of a table and returns information from another column.

Example:

ID Name

101 Ravi

102 Priya

103 Arun

Formula:

=VLOOKUP(102,A2:B4,2,FALSE)

Result:

Priya

XLOOKUP()

A newer and more flexible version of VLOOKUP.

Example:

=XLOOKUP(102,A2:A4,B2:B4)

Result:

Priya

Advantages:

- Can search left or right.
- Easier to use.
- Better error handling.

Common Errors

Error	Meaning
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#DIV/0!	Division by zero
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#VALUE!	Wrong data type
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#NAME?	Function name is incorrect
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#REF!	Invalid cell reference
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#N/A	Value not found
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#NUM!	Invalid numeric value
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